Kimani McLeish

Software Engineer | Data Pipelines and Databases

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PROFESSIONAL SUMMARY

Software engineer with 4 years of hands-on development experience, focused on data pipelines and database-backed systems. Builds scheduled ingestion pipelines (multi-source scraping, validation, deduplication, AI enrichment) and event-driven PostgreSQL/PostGIS backends in Go and TypeScript, and has owned security for 5 production databases on AWS. Strong in SQL, Python and CI/CD; NVIDIA-certified in AI infrastructure and operations (NCA-AIIO).

SKILLS

Databases and Data Engineering: SQL, MySQL (stored procedures), PostgreSQL, PostGIS, H3, MongoDB, Redis, DuckDB, dbt; ETL pipelines, data cleaning and deduplication, scheduled batch jobs, Postgres job queues, Python data-cleaning and import scripts, audit and exception reporting
Languages: Python, SQL, Go, JavaScript/TypeScript, C++, Swift
Cloud and Infrastructure: AWS EC2, Docker, Kubernetes, Nginx, GitHub Actions (CI/CD), Azure App Services, Firebase, Supabase
APIs, Frameworks and Testing: REST APIs, Node.js, Express, React Native (Expo), Next.js, Angular, OpenAI, Apple HealthKit, Fitbit, Mapbox, Socket.io; Jest, Playwright, Postman, UAT

WORK EXPERIENCE

Cybersecurity & Infrastructure Team Lead (Contract) 04/2026 - Present
Sagicor Innovation Lab, Kingston, Jamaica

- Led the team owning security, scaling and maintenance of 14 AWS servers across two regions and 5 production databases, with Docker and Nginx deployments across environments.
- Hardened server and security-group configurations to reduce attack surface; the platform passed an external penetration test by a third-party security firm.
- Built isolated UAT and development environments mirroring production, plus a blue-green deployment pipeline in GitHub Actions that ships updates across regions with zero downtime.
- Presented the platform's security architecture to external stakeholders, including the CTO and IT department heads at the University of Technology, Jamaica.
- Authored architecture and infrastructure documentation adopted by a 7-person team, standardising how servers are provisioned, secured and deployed and reducing deployment errors.

Software Engineer, Internship 12/2025 - Present
Intellibus, Kingston, Jamaica

- Integrated a CMS into the AI Academy website so non-technical staff can publish blog content without a redeploy.
- Translated Figma designs into production-ready components and integrated REST APIs, working with backend teams to keep data flow consistent.

Software Engineer (Contract) 06/2025 - 03/2026
Sagicor Innovation Lab, Kingston, Jamaica

- Consolidated multi-step transaction logic into MySQL stored procedures invoked directly by the API, cutting round-trips and reducing average query response time across high-volume workflows.
- Engineered REST APIs in Node.js and Express for SwiftPay, a financial transaction platform serving 12,000+ university students, covering transfers, balances and JWT-protected endpoints.
- Gathered requirements from stakeholders including RADA and Sagicor department heads across IT, insurance and health, and turned recurring review feedback into product changes.
- Developed CardioScore, a cross-platform health analytics app integrating Apple HealthKit and Fitbit (OAuth2) data, with the full backend and a CI/CD pipeline on AWS EC2; co-built an insurance claim-form system that automated claims processing.
- Authored onboarding documentation and user test-case documents; led user training for SwiftPay, RootIQ and CardioScore.

PROJECTS

Cite: Community Map with Automated Event Ingestion Pipeline
TypeScript, Node, PostgreSQL/PostGIS, Redis, DuckDB, dbt, React Native (Expo)

- Built an automated ingestion pipeline that collects events from 5 web sources every 6 hours, with per-host rate limits, idempotent upserts and cross-source deduplication.

- Wrote a validation layer that skips bad records with a counted reason (no location, vague venue, outside Jamaica), giving every run a data-quality report.
- Added AI enrichment: a vision model (gpt-4o-mini) reads flyers and recovers venues, with strict JSON output, a 0.6 confidence threshold, geocoding fallbacks and a moderator approval queue.
- Seeded a national dataset of emergency and public places from OpenStreetMap (Overpass API) and ran about 12 scheduled jobs across servers, using Redis locks so each job runs on exactly one server.
- Modelled the data as a star schema in DuckDB: 3 fact tables (events, ingestion runs, skipped listings) and 5 dimensions (source, venue, category, skip reason, date), loaded nightly to keep history the app's content expiry deletes.
- Built 12 dbt models with 26 data tests, scheduled with GitHub Actions, powering a dashboard of source yield, skip rate by reason, enrichment success rate and events by category.
- Designed a trust-and-safety layer behind deny-all Postgres row-level security; ~81K lines with 2,000+ automated tests (Jest, Playwright).

Tabi: Location-Based Social App (jointabi.com)
Go, PostgreSQL/PostGIS, H3

- Designed a PostgreSQL/PostGIS schema (138 versioned migrations, row-level security) for a ~96k-line Go backend, using H3 cells so the system stores only users' rough location; ~147 Go test files run against a real Postgres database.
- Modelled gameplay as append-only event ledgers and built transactional batch jobs that turn them into world and regional leaderboards, using window functions and advisory locks.
- Built a Postgres job queue (FOR UPDATE SKIP LOCKED, retries, crashed-worker recovery) running scheduled aggregation, data-retention and notification jobs.
- Built an ETL pipeline that pulls OpenStreetMap venues through the Overpass API, maps raw tags to a venue catalogue and refreshes it every 30 days with retries.

EDUCATION AND CERTIFICATIONS

B.Sc. in Computing
University of Technology, Jamaica, Kingston, Jamaica

Coursework: Database Design, Analysis of Algorithms, Software Engineering, Object-Oriented Programming.
Hackathons: Microsoft Hackathon Jamaica '23/'24 (two-time finalist), Project Blue Mountain Hackathon, Intellibus Hackathon (Squire).

NVIDIA-Certified Associate: AI Infrastructure and Operations (NCA-AIIO), NVIDIA | 08/2026 - 08/2028

LEADERSHIP

President, Future Developers Club, University of Technology, Jamaica. Member, IEEE and SAPNA Clubs: robotics and software innovation projects.